

Short communication

Comparison of atmospheric polycyclic aromatic hydrocarbon levels in three urban areas in Lebanon

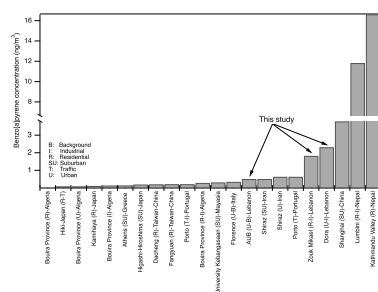
Rima Baalbaki^a, Julie Nassar^a, Samer Salloum^b, Alan L. Shihadeh^b, Issam Lakkis^b, Najat A. Saliba^{a,*}

^a Department of Chemistry, Faculty of Arts and Sciences, American University of Beirut, Beirut, 1107 2020, Lebanon

^b Department of Mechanical Engineering, Maroun Semaan Faculty of Engineering and Architecture, American University of Beirut, Beirut, 1107 2020, Lebanon



GRAPHICAL ABSTRACT



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ABSTRACT

Introduction: Reports on the levels and sources of PAHs in the Middle Eastern region are very scarce and mostly outdated. This study reports for the first time the ambient levels of PAHs during summer and winter at three coastal locations in Lebanon. Comparison between PAH levels in the three locations is established and the contribution of major emission sources to PAHs is evaluated.

Materials and methods: Gas and particle phase PAHs were collected using a high-volume sampler fitted with a quartz filter and a polyurethane foam (PUF) holder at three coastal sites (American University of Beirut, Dora, and Zouk Mikael). The collected PAHs were then extracted and analyzed by gas chromatography-mass spectrometry (GC-MS). The dispersion of pollutants in the emissions from a local power plant was evaluated using The Air Pollution Model (TAPM), coupled with field measurements to quantify the BaP level in two nearby locations and one background location. One-way ANOVA tests and coefficients of divergence (COD) between paired sites were used for data analysis.

Results: High heterogeneity was detected in the temporal and spatial variations of the PAH profiles and concentrations at the three coastal urban sites. Higher PAH levels were measured in winter than in summer at AUB and Dora, but not at Zouk Mikael. Major sources of pollution were identified as traffic, diesel engines, road dust, and oil fuel. Diesel engines are used as local sources of electricity during rotating blackout and oil fuel are used to run the inadequately maintained local power plant. The BaP levels measured near the power plant were approximately 10 times higher than those at the background site.

Conclusions: The annual average BaP concentrations of 2.07 ng m^{-3} and 2.9 ng m^{-3} at Zouk Mikael and Dora,

* Corresponding author.

E-mail address: ns30@aub.edu.lb (N.A. Saliba).

respectively, exceeded the European Union (EU) air quality standard of 1 ng m^{-3} . Owing to the lax governmental regulations, the BaP levels in Lebanon are higher by 60%–99% than those in most cities around the world.

1. Introduction

The association between elevated risks of diseases and exposure to air pollution has been established by several time-series cohort studies (Brook, 2008; Pope and Dockery, 2006; Pope et al., 2006; Simkhovich et al., 2008). Of special interest among the air pollutants are polycyclic aromatic hydrocarbons (PAHs), which are semi-volatile organic compounds with two to more fused aromatic rings. These compounds have attracted much research attention ever since benzo[a]pyrene (BaP), a five-membered ring PAH, was first isolated from coal tar in 1933 and identified as a potent chemical carcinogen (Cook et al., 1933). PAHs are ubiquitous and highly mobile in the ambient atmosphere. Their accumulation in the atmosphere result from the PAH long-range transport concentrations and the concentrations of local petrogenic and pyrogenic emissions. Hence, PAH profiles are expected to differ quantitatively and qualitatively between cities and within cities if the sampled sites are exposed to different local emission sources and meteorological conditions (Li et al., 2016; Snyder et al., 2010).

Studies reporting the levels of airborne PAHs in Middle Eastern countries are scarce and Lebanon is no exception (Alghamdi et al., 2015; Mastral et al., 2003). However, indications of high PAH levels can be deduced from several studies on particulate matter (PM) levels and their chemical composition (Daher et al., 2013, 2014). Over the past decade, ambient PM levels in the greater Beirut area of Lebanon have been found to consistently exceed the WHO standard of $10 \mu\text{g}/\text{m}^3$ (Daher et al., 2013; Kouyoumdjian and Saliba, 2005; Massoud et al., 2011; Nakhle et al., 2015; Saliba, 2004) and the daily PM_{10} and $\text{PM}_{2.5}$ concentrations have been associated with increasing number of respiratory and cardiovascular emergency hospital admissions (Nakhle et al., 2015). PM-bound PAHs near a major road in Beirut were found at concentrations up to four times higher than the concentrations measured at the I-710 freeway in Los Angeles (Daher et al., 2013). Although the sources of PAHs at this site can be linked to gasoline and diesel motor vehicles, it is not clear how much these levels differ from the ambient concentrations and whether other emission sources, such as heavy fuel oil power plants and private diesel-powered electric generators, contribute to the increase in health hazards due to PM exposure. These multiple uncontrolled point sources result in highly

variable ambient PM and PAH levels, which can only be characterized accurately at the city level.

This study reports for the first time the ambient levels of PAHs during summer and winter at three coastal locations in Lebanon. Comparison of atmospheric PAH levels are assessed and the contribution of major emission sources is evaluated. Of particular concern is the extent at which the power plant in Zouk Mikael affected the PAH levels in the adjacent communities. This was addressed based on simulation studies using The Air Pollution transport Model (TAPM, (Hurley et al., 2005)) and measurements of BaP levels.

2. Materials and methods

2.1. Study area

Ambient PAH measurements were conducted at three coastal locations in Lebanon – AUB, Dora, and Zouk Mikael (Fig. 1) during winter (January–March) and summer (June–July) in the year 2015. The locations include an urban background site (the American University of Beirut (AUB)), an industrial area with a major commuter highway (Dora), and a site affected by a major heavy fuel oil power plant (Zouk Mikael). The AUB site is located on the rooftop of an academic building of AUB situated in the Ras Beirut neighborhood. It is surrounded by trees on all sides and is away from any major street by at least 200 m. The Dora sampling site was located on the third-floor balcony of a retail/commerce building and is one block away from a major commuter highway that remains congested throughout the day. Dora is a mixed residential and light industrial area. The Zouk Mikael sampler was located on the roof of the Zouk Mikael municipality building, located in the heart of a residential community north of Beirut. The building is approximately 2 km east of a major regional power plant.

2.2. Sampling and extraction of gas and particle-bound PAHs

Twenty-four hour air samples were collected in each season in AUB, Dora, and Zouk Mikael ($n = 36$, with 6 samples/site/season) consecutively using a high-volume sampler (HVS) (DH77, DIGITEL Elektronik AG, Hegnau, Switzerland) operating at a flow rate of 500 L/

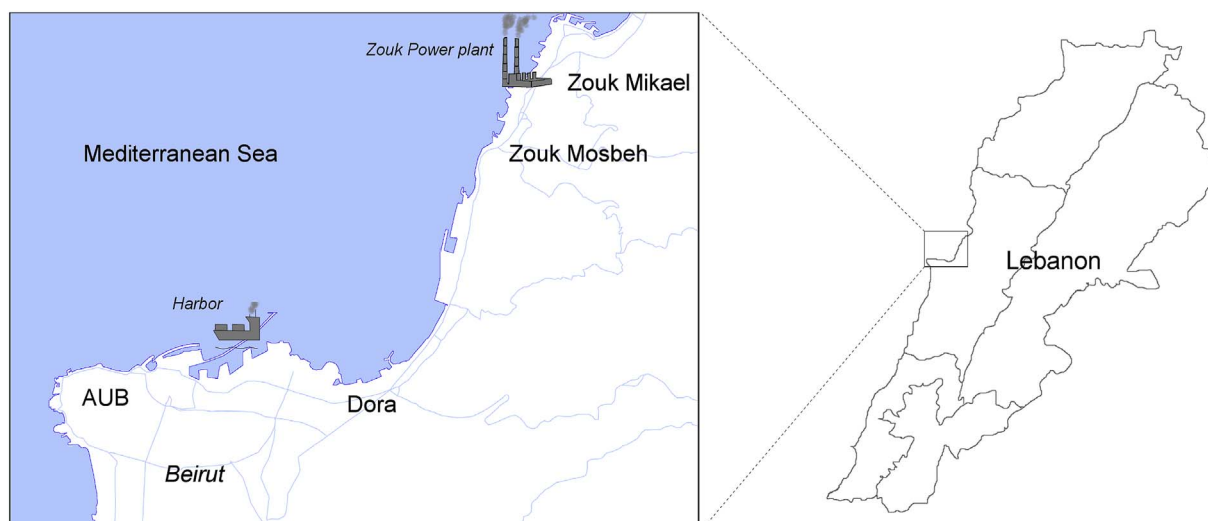


Fig. 1. Area of study showing the sampling sites – AUB, Dora, Zouk Mikael and Zouk Mosbeh.

min and equipped with a PM₁₀ impactor. Quartz fiber filters (Whatman, 150 mm), pre-baked at 400 °C for 5 h, were used to collect particulate PAHs. Polyurethane foams (PUF), Soxhlet-cleaned for 12 h prior to sampling, were used to collect gaseous PAHs. Post sampling, both the PUF and filter were wrapped in baked aluminum foils and stored at –20 °C until analysis. Both the filter and PUF were spiked with 2 µg internal standards (deuterated PAHs). They were extracted together using a mixture of 300 mL of diethyl ether and hexane (10:90 v/v ratio) in a Soxhlet extractor adjusted at an evaporation-condensation rate of 2 drops/s for 18 h. The extract was then rotary concentrated to 10 mL and reduced further under nitrogen flow to 1 mL. The 1 mL extract was cleaned on a HyperSep Silica SPE (solid phase extraction) cartridge using 10 mL of hexane and then re-concentrated to 1 mL under nitrogen flow prior to analysis.

Further, low volume samplers (LVS) were used to collect 24-h PM₁₀ samples on 47 mm quartz filters at three sites identified by TAPM (section 3.3). These filters were extracted by 2 h of sonication in 8 mL of toluene followed by 30 min of shaking at 250 rpm. The filters were then removed and the sample volume was reduced to 1 mL. Subsequently,

the extract was cleaned on a SPE and concentrated to 200 µL.

2.3. Gas chromatography-mass spectrometry (GC-MS) analysis

The extracted samples were analyzed on a Thermo Trace GC Ultra chromatographer equipped with an ITQ-900 MS detector. Chromatographic separation was carried out on a Rtx-5MS column (60 m × 0.25 µm film thickness × 0.25 mm film ID) using helium (1 mL/min) as the carrier gas. A splitless injection volume of 1 µL was used and the injector temperature was set at 280 °C. The temperature program of the oven was as follows – 80 °C for 4 min, heating at 10 °C/min to 170 °C (hold for 1 min), heating at 3 °C/min to 180 °C, heating at 10 °C/min to 270 °C, and heating at 3 °C/min to 300 °C (hold for 3 min). The mass spectrometer was operated in the electron impact ionization mode with an ionizing energy of 70 eV over an m/z scanning range of 50–350 at an ion source temperature of 250 °C. Quantification was done in selected ion current profiles using a seven-point calibration curve for PAH standards (0.05 and 10 ppm) and deuterated PAH standards of 2 ppm (d-acenaphthylene, d-phenanthrene, d-chrysene, and d-

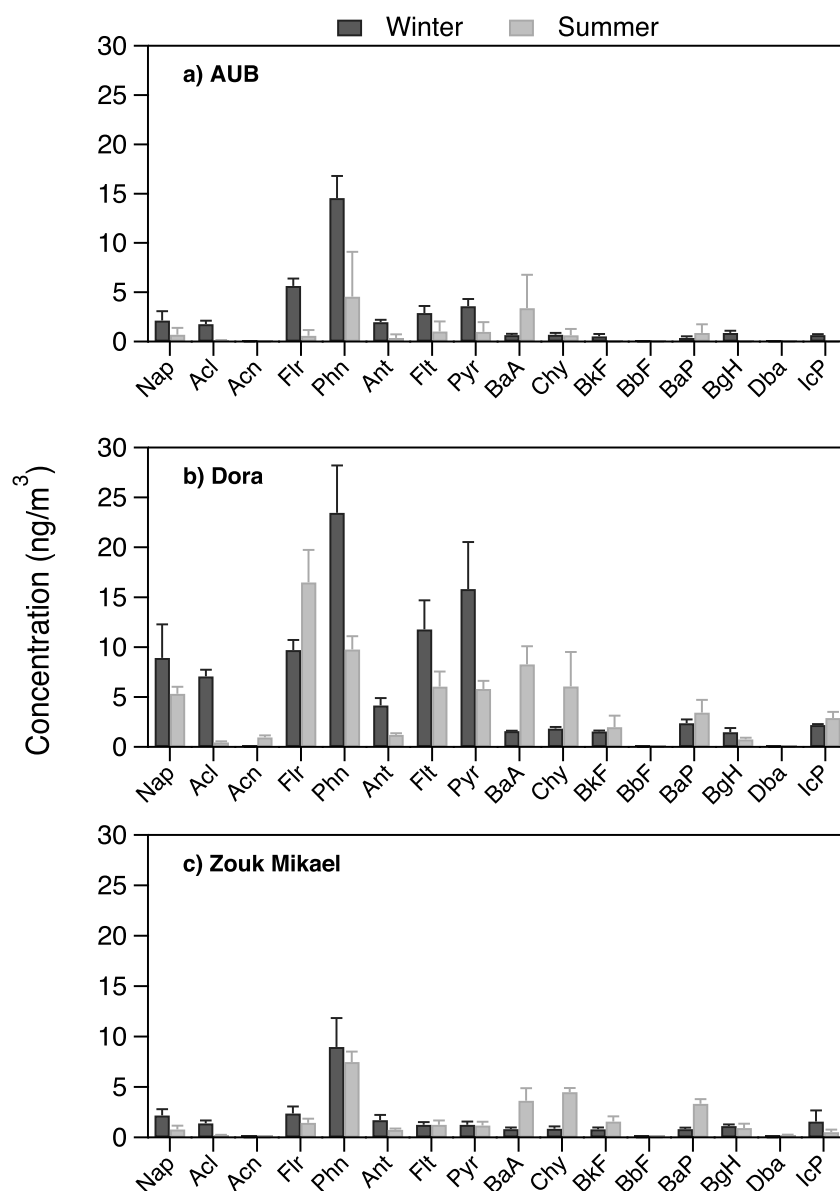


Fig. 2. Average winter and summer concentrations of 16 PAHs at AUB (a), Dora (b), and Zouk Mikael (c). The error bars represent standard deviations from 6 samples. Nap (naphthalene), Acl (acenaphthylene), Ant (anthracene), Flt (fluoranthene), Pyr (pyrene), BaA (benzo[a]anthracene), Chy (chrysene), BkF (benzo[k]fluoranthene), BbF (benzo[b]fluoranthene), BaP (benzo[a]pyrene), BgH (benzo[g,h,i]perylene), DbA (dibenz[a,h]anthracene), and ICP (indeno[1,2,3-cd]pyrene).

perylene) dissolved in either hexane or toluene, depending on the type of samples being extracted (HVS or LVS). The correlation factor of the calibration curves (R^2) ranged between 0.995 and 0.999. All reported values were corrected for blank filters, which were extracted with every batch of analysis. The overall recovery of each PAH ranged between 65 and 128%. The repeatability was determined by running four consecutive samples of 2 ppm each during the same day. Results showed that the relative standard deviation ranged between 1.4 and 8.2%. The estimated limit of detection (LOD) ranged between 8 and 38 pg/m^3 for HVS samples and between 90 and 440 pg/m^3 for LVS samples. The LOQ values for HVS ranged between 38 and 125 pg/m^3 .

2.4. Zouk Mikael power plant

The effect of emissions from the Zouk Mikael power plant on the nearby areas was evaluated based on the data on fuel consumption rate, stack dimensions, and exhaust temperature. PM emissions were estimated assuming an emission factor of 0.71 g TSP (Total Suspended Particles) per kg fuel burned and a $\text{PM}_{10}/\text{TSP}$ ratio of 0.5 (Goldstein and Siegmund, 1976). This data was included in TAPM – a software to predict pollutant transport – taking into account the atmospheric chemical reactions and species advection by weather flow field patterns (Hurley et al., 2005). Additional information on TAPM theory and run parameters can be found in the supplementary information.

2.5. Statistical analysis

One-way ANOVA tests were conducted to compare the PAH concentrations at different sites across different seasons. Coefficients of divergences (COD) between the paired sites (AUB-Dora, AUB-Zouk Mikael, and Zouk Mikael-Dora) were used to assess their degree of uniformity as well as the PAH spatial variability. The COD is then computed as

$$\text{COD} = \sqrt{\frac{1}{n} \sum_{i=1}^n \left(\frac{x_i - y_i}{x_i + y_i} \right)^2}$$

where x and y are two measured concentrations at different sites on the i th day and n is the total number of observations. High COD levels imply heterogeneity between the sites while a low value indicates similarities between the sites. Based on literature review (Krudysz et al., 2008, 2009; Miller et al., 2010; Moore et al., 2009; Sarnat et al., 2010; Wilson et al., 2005), when the values of COD are above 0.2, it is concluded that there are variations between the sites (Wilson et al., 2005).

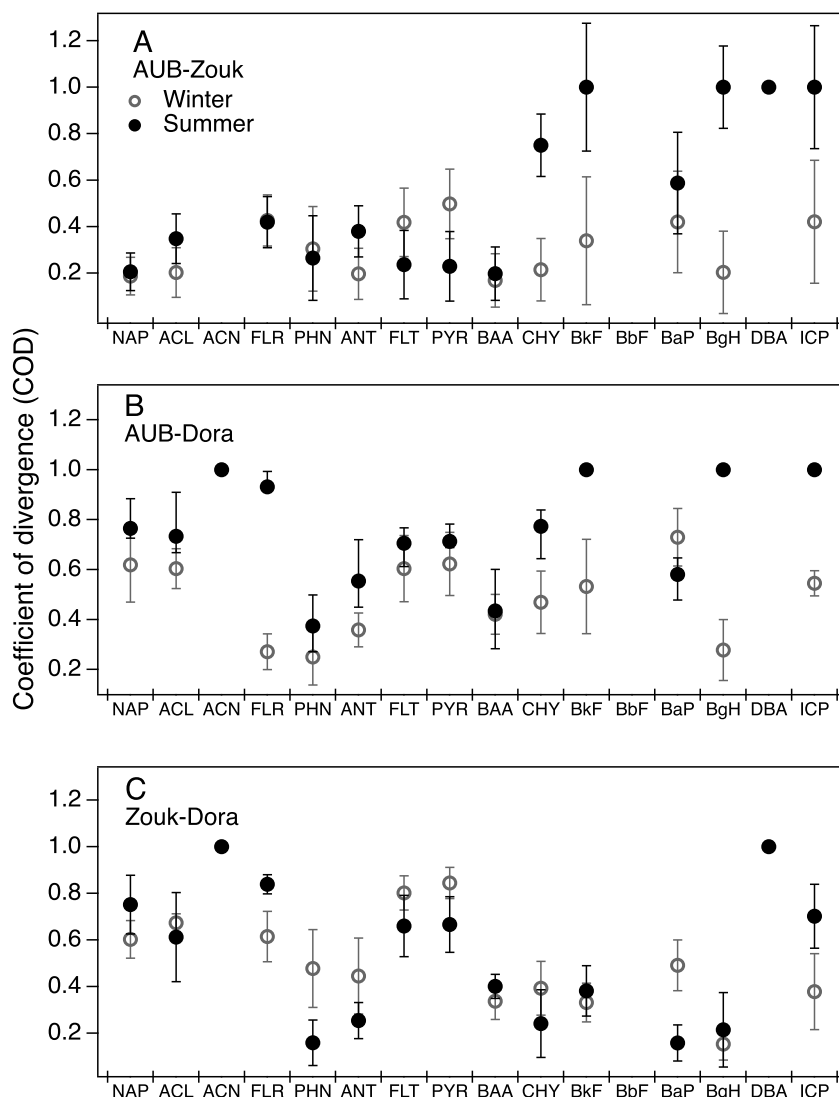


Fig. 3. Coefficient of divergence (COD) plotted for 16 PAHs for each pair of sites – (A) AUB-Zouk Mikael, (B) AUB-Dora, and (C) Zouk Mikael-Dora.

3. Results

3.1. Levels and seasonal variations of atmospheric PAHs at the three sampling sites

The levels of and variations in PAH concentrations at each of the three sampling sites during winter and summer are shown in Fig. 2. The average PAH concentrations at AUB registered (36.30 ± 2.8) ng m^{-3} in winter and (13.15 ± 1.18) ng m^{-3} in summer. The levels of PAHs in Zouk Mikael were (25.08 ± 3.31) ng m^{-3} and (27.65 ± 2.05) ng m^{-3} in winter and summer, respectively. The highest PAH average concentrations were recorded in Dora at (91.88 ± 8.18) ng m^{-3} in winter and (69.43 ± 5.86) ng m^{-3} in summer. This difference in PAH concentrations at different sites was statistically significant as determined by one-way ANOVA ($F(2,30) = 89.815$, $p < 0.001$).

At the AUB and Dora sites, higher PAH concentrations were noted in winter than in summer. The difference between summer and winter PAH concentrations was found to be statistically significant at both sites (AUB: $F(1,9) = 110.945$, $p = 0.000$); Dora: $F(1,9) = 23.506$, $p = 0.01$). In contrast, the PAH concentrations in Zouk Mikael did not differ across seasons ($F(1,9) = 1.058$, $p = 0.33$). On examining the BaP levels at each site, it is noticed that in summer, Zouk Mikael ((3.32 ± 0.46) ng m^{-3}) and Dora ((3.43 ± 1.28) ng m^{-3}) exhibited higher concentrations than in winter Zouk Mikael ((0.82 ± 0.14) ng m^{-3}) and Dora ((2.37 ± 0.37) ng m^{-3}).

3.2. Comparison of atmospheric PAH levels at the sampling sites

For comparison purposes, the profiles of COD (Fig. 3) and individual PAH normalized concentrations to the total PAH concentration were used to determine PAH similarities at the three sites (Fig. 4). Most values of COD are above 0.2 concluding that there are variations between the sites. The contributions of individual PAHs to the total sum varied. In particular, it is noted that, in winter, the 2- and 3- benzene ring PAHs, such as fluorene (Flr) and phenanthrene (Phn) were the highest in AUB followed by Zouk Mikael and Dora. Dora exhibited the highest

levels of fluoranthene (Flt) (3-benzene rings) and pyrene (Pyr) (4-benzene rings) while Zouk Mikael showed the highest levels of BaP, benzo[ghi]perylene (BgH) and indeno[1,2,3-cd]pyrene (IcP) (5-benzene rings). The differences in the site profiles were even more enhanced in summer where high contributions were Phn, and BaA for AUB, Flr and Flt for Dora, and chrysene (Chy) and BaP for Zouk Mikael.

3.3. The effect of Zouk power plant on the Zouk area

The high levels of BaP and BgH observed in Zouk Mikael in both winter and summer highlight the role of the local power plant in increasing the PAH emissions. The contribution of this local source to PAH levels at the site was assessed using TAPM and a field measurement of particle-bound BaP. TAPM tracks PM_{10} concentrations based on a simple tracer method in the selected region using the accessible meteorological data of the year 2014. The simulation results showed that the power plant contributed to the pollution in the surrounding area to a large extent. According to TAPM, the polluted zones are most likely situated to the east and southeast of the power plant due to the prevailing wind direction (Fig. S1). Consequently, two distinct zones most affected by the power plant emissions were identified. The first zone is situated to the east of the power plant at a distance of 2–4.7 km (Zouk Mikael) and the second zone is situated to the southeast of the power plant at a distance of around 4.8 km (Zouk Mosbeh). Accordingly, the three simultaneous measurements of BaP which were conducted in Zouk Mikael, Zouk Mosbeh, and at a background site situated in Beirut (AUB) showed BaP concentrations in Zouk Mikael 4.5 and 10 times higher than the BaP concentrations in Zouk Mosbeh and AUB, respectively (Fig. 5).

4. Discussion

The PAH levels at the three coastal sites (AUB, Dora, and Zouk Mikael) were determined using an HVS coupled with a GC-MS. Statistically significant temporal and spatial differences were found within (intra-) and between (inter-) different sites. It was noticed that

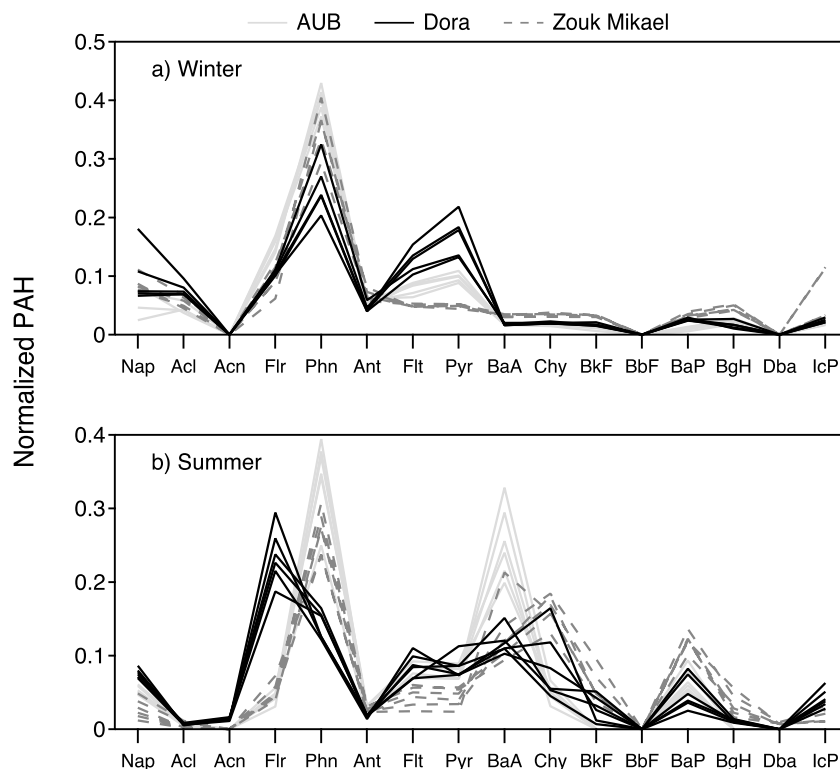


Fig. 4. PAH concentrations normalized to total PAH concentrations measured during winter (a) and summer (b) at AUB, Dora and Zouk Mikael sites.



Fig. 5. Box plots of BaP concentrations measured at AUB, Zouk Mosbeh and Zouk Mikael. Whisker bottom, box bottom, box top and whisker top represent 10, 25, 75 and 100 concentration percentiles.

the PAH levels in winter were higher than those measured in summer at AUB and Dora. These results are consistent with previous studies which reported winter to summer ratios in the range of 1.5–4.5 (Baek et al., 1991; Caricchia et al., 1999; Harrison et al., 1996; Menichini, 1992). The higher concentrations in winter have been attributed to several

factors including the higher frequency of temperature inversions, increased usage of PAH emitting sources, such as domestic heating, and slower PAH photodegradation reactions (Keyte et al., 2013; Kim et al., 2009). In Zouk Mikael, the 300% increase in BaP in the summer is likely due to a greater need for electricity during the hot season. In fact, the Lebanese Ministry of Energy and Water reported an increase in electricity consumption in the summer of 2009 of 63% over the annual average capacity (Bassil, 2010).

The COD values for most PAHs were higher than 0.2, indicating that PAH heterogeneity mostly originated from the variations in the emission load and source type. Because the sites are less than 10 km apart, it is unlikely that the fates of the PAHs, due to atmospheric transformations at the three sites, were different and so heterogeneity is attributed to intra-urban variations. Intra-urban spatial variability has been emphasized in several studies (Chakraborty et al., 2010; Li et al., 2016; Snyder et al., 2010; Totten et al., 2004; Yadav et al., 2017). One example is the 36-site mobile sampling campaign conducted during the summer and winter of 2013 in Pittsburgh, PA; it indicated significant intra-urban spatial variability in fine particle composition (Li et al., 2016). Another example is the spatial variability of carbonaceous aerosols and associated source tracers in Detroit, Michigan and Cleveland, Ohio, where a comparison of 24 h measurements of fine carbonaceous aerosols revealed substantial intra-urban variability, which is

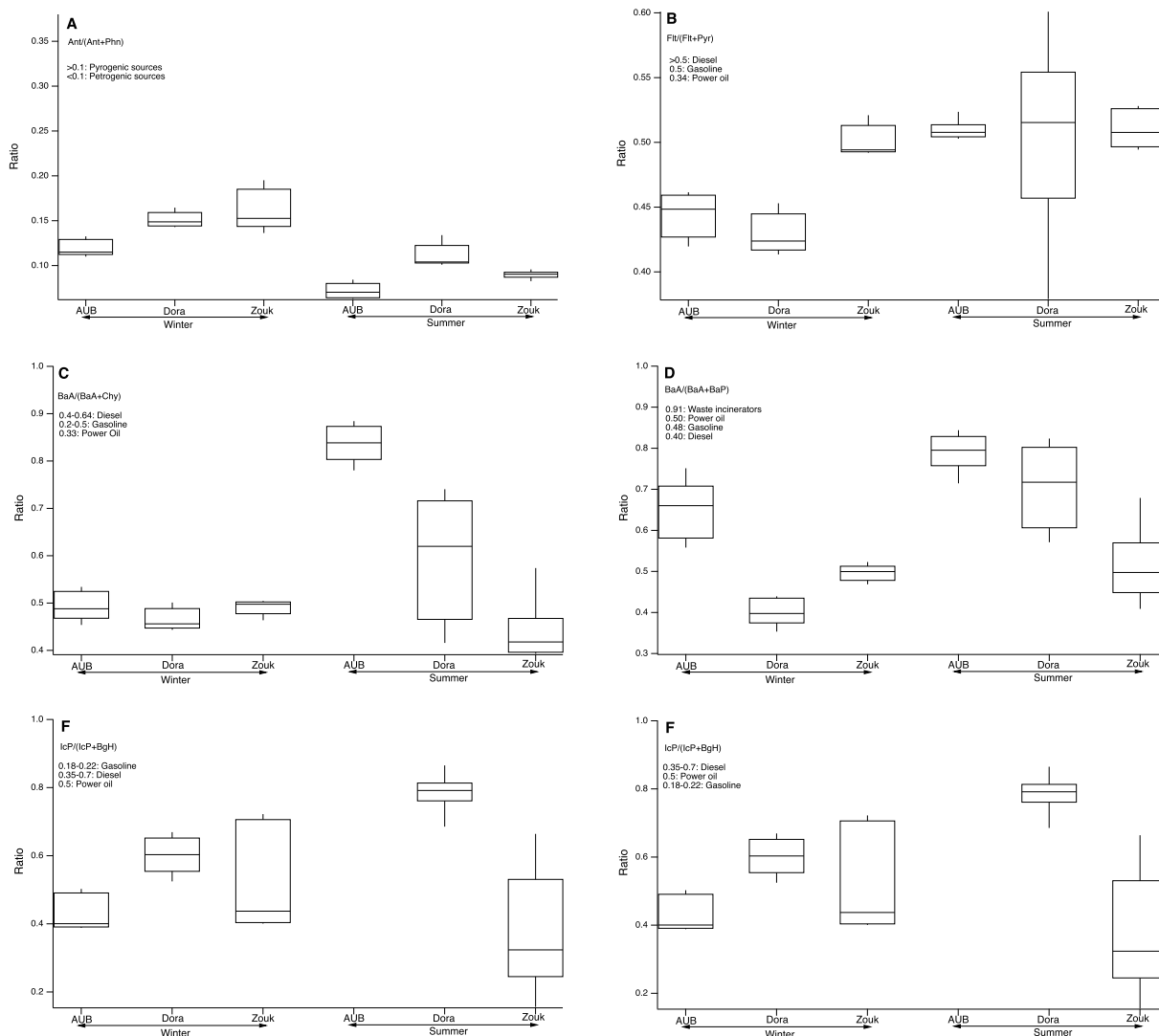


Fig. 6. A–F. Ratios of frequently identified PAH isomers Ant/(Ant + Phn) (A), Flt/(Flt + Pyr) (B), BaA/(BaA + Chy) (C), BaA/(BaA + BaP) (D), Pyr/BaP (E), and IcP/(IcP + Bghi) (F) for each of the sample location in Winter and Summer. Assigned sources to isomer ratios are extracted from Bian et al. (2016) and Cecinato et al. (2014), Tobiszewski and Namiesnik (2012) and Ravindra et al. (2008).

Table 1
Benzo[a]pyrene (BaP) concentrations (ng/m³) reported in different regions.

Date of sampling	Location	BaP	Sampling phase	Reference
May 2014 to Jan 2015	Dacheng-Taiwan-China (residential)	0.19 ± 0.11	PM _{2.5}	Chen et al., 2016a,b
	Fangyuan-Taiwan-China (residential)	0.20 ± 0.13		
Apr 2013 to Mar 2014	Lumbini-Nepal (rural affected by industry zones)	11.8 ± 9.32	TSP	Chen et al., 2016a,b
Aug 2013 to Jul 2014	Kamihaya-Japan (not populated)	0.1 ± 0.13	TSP	Jadoon and Sakugawa 2016
	Hiki-Japan (rural-Traffic)	0.08 ± 0.08		
	Higashi-Hiroshima-Japan (semi-urban)	0.18 ± 0.13		
Aug to Dec 2015	Shanghai-China (sub-urban)	3.77 ± 1.46	TSP ^a	Wang et al., 2016
Jun–Sep 2013 & Jan–Feb 2014	University Kebangsaan-Malaysia (semi-urban)	0.3	PM _{2.5}	Khan et al., 2015
April 2013–March 2014	Kathmandu Valley-Nepal (agricultural–residential area)	16.6 ± 15.2	TSP	Chen et al., 2015
2014	Metropolitan Area of Porto-Portugal (traffic-industrial)	0.2	PM ₁₀ and	Albuquerque et al., 2016
Mar–May 2014	Bouira province-Algeria (industrial)	0.12 ± 0.06	PM ₁₀	Khedidji et al., 2017
	Bouira province-Algeria (residential)	0.03 ± 0.01		
	Bouira province-Algeria (urban)	0.08		
	Bouira province Algeria (residential near cement plant)	0.27 ± 0.55		
spring of 2015	Paramont, Shiraz-Iran (sub-urban)	0.73 ± 0.88	PM ₁₀	Shahsavani et al., 2017
	Sadra, Shiraz-Iran (urban)	0.43 ± 0.61		
2013 to February 2014	Oporto-Portugal (traffic site)	0.62	PM _{2.5}	Alves et al., 2017
	Florence-Italy (urban background)	0.33		
	Athens-Greece (sub-urban)	0.13		
Winter and summer 2015	AUB (urban background)	0.49 ± 0.26	PM ₁₀	This study
	Zouk Mikael (residential affected by power plant)	1.81 ± 1.09		
	Dora (traffic industrial)	2.30 ± 1.00		

^a Sum of 5 stages of Anderson impactor.

consistent with the influence of local sources (Snyder et al., 2010).

The calculations based on comparing ratios of pairs of most frequently identified PAH isomers help to characterize the origins of PAHs in ambient air (Fig. 6A–F). Consequently, the most common ratios of Ant/(Ant + Phn), Flt/(Flt + Pyr), BaA/(BaA + Chy), BaA/(BaA + BaP), Pyr/BaP, and ICP/(ICP + BgH) were used to investigate the sources contributing to PAH concentrations in air (Bian et al., 2016; Cecinato et al., 2014; Ravindra et al., 2008; Tobiszewski and Namiesnik, 2012). The study of the diagnostic ratios suggested a strong influence of traffic, diesel burning, dust from paved roads, oil fuelled combustion for power production, and waste incinerators as major sources of PAH concentrations in the atmosphere. Variations in the ratios between the three sites reflected differences in the source predominance in each location. Diesel engine emissions in Beirut originate from two sources, traffic and diesel engines used as local sources of electricity. Lebanon faces infrastructure shortage, due to which the government is incapable of maintaining power plants properly and producing electricity sufficient for the market's needs. This led the Lebanese government to institute rotating blackout regimens and in response, local neighborhoods produce their own power using diesel generators. Emissions from the power plant and the extent to which they affect the surrounding areas were identified using TAPM and a field measurement of particle-bound BaP. The BaP concentration measured in a location to the east of the power plant was 4.5 times higher than the levels measured at a distance of around 4.8 km south-east of the power plant.

In comparison to other cities in developed and developing countries, Table 1 summarizes the BaP concentrations reported in various cities in the past three years, (Albuquerque et al., 2016; Alves et al., 2017; Chen et al., 2015, 2016a, 2016b; Jadoon and Sakugawa, 2016; Khan et al., 2015; Khedidji et al., 2017; Shahsavani et al., 2017; Wang et al., 2016). The BaP levels reported in this study are 60%–99% higher than those in most cities around the world. BaP concentrations measured at the residential-industrial site (Dora) and the residential site affected by the power plant (Zouk Mikael) were 2.3 and 1.8 times higher than the EU standard (EEA, 2013, 2015) while that measured at the urban background site (AUB) was lower than the standard.

5. Conclusions

This study reports the sum of gaseous and particle PAH

concentration in three urban locations along the coast of Lebanon. PAHs differed between sites and between winter and summer within the same site in AUB and Dora. A comparison of the ratios of pairs of most frequently identified PAH isomers showed that the main sources of emissions are traffic, diesel generators, road dust, oil fuel, and waste incinerators. Comparing to other cities, BaP concentrations are 60%–99% higher than most cities around the world. In fact BaP levels exceeded the EU standard (1 ng/m³) in two out of the three studied residential areas.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.atmosenv.2018.02.028>.

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